

Mahmood Muwafi

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PROFESSIONAL SUMMARY

Ambitious **Junior Software Engineer** and final-year student with a focus on **Computer Vision (CV)** and **AI-driven applications**. Proven expertise in building enterprise-grade systems through an **Infrastructure Networking internship at UNEC** and a peer-reviewed **IEEE publication** on advanced person profiling. Proficient in **Python (TensorFlow/Keras), React, and MediaPipe**, with a dedicated interest in developing accessible health-tech and high-security analytics.

EDUCATION

Bsc. Computer Science, Specialized in Artificial intelligence

GPA – 3.8

The British university in Dubai

2026

Dubai, United Arab Emirates

INTERNSHIP EXPERIENCE

Infrastructure Networking and AI intern

UNEC (United National Engineering & Construction)

Employer: Ali Jechi

Apr 2025 – Aug 2025

Dubai, United Arab Emirates

- **Developed and deployed image recognition and human detection models** using **PyTorch and Keras**, achieving an **85% accuracy rate** in identifying personnel within complex construction environments.
- **Engineered a prototype security solution** for the DEWA Headquarters 8th floor by integrating **AI models with Arduino-based hardware** to automate employee access control and door-lock mechanisms.
- **Assisted a Senior CCIE Engineer** in the configuration and deployment of enterprise network switches via **SSH**, maintaining high-value infrastructure for a major utility headquarters.
- **Audited and documented technical blueprints** for CCTV wiring across multiple floors using **Excel**, ensuring that physical installations aligned with engineering specifications and workshop documentation.
- **Collaborated with the Senior Smart Engineer** on daily site inspections and technical walkthroughs to monitor the progress of "Smart Building" infrastructure components.

PROJECTS

Physio Vision (physiovision.app) | Python, MediaPipe, React, SQLite, PyQt5

Feb 2026 – July 2026

- **Architected a full-stack rehabilitation platform** designed for the elderly and physically impaired, utilizing **MediaPipe Pose** and **LSTM neural networks** to achieve **90% accuracy** in movement analysis.
- **Built a medical-grade desktop GUI** using **PyQt5** that features holographic **AR positioning guides** and asynchronous audio coaching for real-time form correction.
- **Developed a fully functional web portal** in **React** with an **SQLite** backend, allowing users to register, download local software, and track session history via encrypted cloud sync.
- **Implemented automated biomechanical diagnostics** for exercises like squats and pushups, providing specific feedback on joint angles and compensatory movements (e.g., back rounding).

UrbanSight Analytics | Python, ONNX, Keras, Scikit-Learn

Nov 2025 – Jan 2026

- **Published and presented** a person profiling system at the **17th Student Research Conference (SRC 2026)** at Zayed University, accepted into the **IEEE Xplore** database.
- **Developed six optimized AI models** for real-time analysis of security footage, automating the detection of age, ethnicity, and clothing attributes for airport safety.
- **Engineered a K-Nearest Neighbors (KNN)** color detection algorithm and utilized **ONNX inference sessions** to achieve lightning-invariant accuracy across diverse surveillance environments.

TECHNICAL SKILLS

- **Languages** | Python (Expert), JavaScript (ES6+), SQL (SQLite), HTML5, CSS3. Go / C# (Currently Learning).
- **AI & Computer Vision** | MediaPipe, PyTorch, TensorFlow, Keras, ONNX Runtime, OpenCV, Scikit-Learn.
- **Web & Backend** | FastAPI, React, RESTful APIs, Uvicorn, Pydantic v2.
- **Security & Auth** | JWT (HS256), Bcrypt (Adaptive Hashing), OAuth 2.0, Rate Limiting (SlowAPI), Input Validation.
- **Infrastructure & Tools** | Raspberry Pi 5, Git, Docker, PyQt5, JIRA, SSH, Linux (Debian/Ubuntu), Visual Studio Code., kaggle
- **Methodologies** | Iterative Development, Full-Stack Development, Peer-Reviewed Research (IEEE Standards).